

National Textile University, Faisalabad



Department of Computer Science

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Assignment	1st
Course Name	Coal
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Computer Organization and Assembly Language (CSC-2075)

BSAI4th SP2026

Assignment 1

Part A: Short Questions

1. Define the following data types: **BYTE**, **WORD**, **DWORD**, **QWORD**.

A. These simply represent how much memory a variable occupies.

- **BYTE** = 8 bits = 1 byte
- **WORD** = 16 bits = 2 bytes
- **DWORD** = 32 bits = 4 bytes
- **QWORD** = 64 bits = 8 bytes

2. What is the difference between **BYTE** and **SBYTE**?

A.

BYTE	SBYTE
• It is unsigned value. • It can store only positive values (0-255).	• It is signed value. • It can store both negative and positive values (-128 to +127).

3. What is the maximum value that can be stored in: (a) **BYTE** (b) **WORD** (c) **DWORD**?

A. The maximum value occurs when all bits are 1.

- **BYTE** = 255 (FFh)
- **WORD** = 65535 (FFFFh)
- **DWORD** = 4294967295 (FFFFFFFFh)

4. What is Little Endian order? Explain with example.

A. Little Endian is a way of storing data in memory where the least significant byte (LSB) is stored first.

Example:

If we have the value 12345678h it will be stored in memory as:

78h
56h
34h
12h

So, the bytes are stored in reverse order like smallest part first.

5. What is the purpose of the DUP operator?

A. The DUP operator is used to allocate multiple memory locations with the same initial value.

For example:

If we write 5 DUP(0) it creates 5 memory locations and each one is initialized to 0.

It helps reduce repetition in code.

6. What is the difference between .data and .code? segment?

A. In simple words:

- .data stores data by declaring and initializing the variables.
- .code runs the program contains the instructions that CPU executes.

9. How can we calculate the size of a WORD array?

A. As we know a WORD takes 2 bytes lets calculate the total size as:

$$\text{Size} = \text{Number of elements} \times 2$$

For example:

If there are 5 WORD elements:

$$5 \times 2 = 10 \text{ bytes}$$

10. What is the use of the TYPE operator?

A. The TYPE operator returns the size in bytes of one element of a variable.

For example:

- TYPE of Byte = 1
- TYPE of Word = 2
- TYPE of Dword = 4

11. What is the use of LENGTHOF and SIZEOF operators?

A. Formula:

$$\text{SIZEOF} = \text{LENGTHOF} \times \text{TYPE}$$

- **LENGTHOF** returns the number of elements in an array.
- **SIZEOF** returns the total number of bytes used by the variable.

12. What are the three operand types in assembly language?

A. There are three main operand types:

1. Register operand (like AX, BX)
2. Immediate operand (a constant value like 5 or 10h)
3. Memory operand (a variable name)

13. Why is the following invalid? `mov ax, myByte`

A. This instruction is invalid because **ax** is a 16-bit register while **myByte** is an 8-bit variable.

In the MOV instruction both operands must be of the same size. Since their sizes do not match that's why the instruction is not allowed.

14. What are the rules of the MOV instruction?

A. The MOV instruction is used to transfer data from one location to another. For this there are some rules which ensure proper data movement without ambiguity that some important rules are:

- Both operands must be of the same size (for example: 8-bit to 8-bit, 16-bit to 16-bit).
- Memory-to-memory transfer is not allowed.
- The destination operand cannot be an immediate value.
- Immediate values can only be moved into registers or memory.

15. What is the difference between MOVZX and MOVSX?

A. Both instructions move data from a smaller register to a larger register, but they handle

extra bits differently:

- **MOVZX** (Move with Zero Extension) fills the extra higher bits with zeros. MOVZX is used for unsigned values.
- **MOVSX** (Move with Sign Extension) copies the sign bit (most significant bit) into the extra higher bits. MOVSX is used for signed values.

16. When is the Zero Flag set?

A. The Zero Flag is set $ZF = 1$ when the result of an operation is zero.

For example:

If we subtract 5 from 5, the result is 0, so the Zero Flag becomes 1.

17. When is the Sign Flag set?

A. The Sign Flag is set when the result of an operation is negative.

It checks the most significant bit (MSB):

- If $MSB = 1$ **then**; number is negative and $SF = 1$.
- If $MSB = 0$ **then**; number is positive and $SF = 0$

18. When is the Carry Flag set?

A. The Carry Flag is set when there is an unsigned overflow.

For example:

If we add $255 + 1$ in an 8-bit register the result exceeds the maximum limit so CF becomes 1.

It indicates that a carry was generated out of the most significant bit.

19. When is the Overflow Flag set?

A. The Overflow Flag is set when there is a signed overflow.

This happens when:

- Adding two positive numbers gives a negative result.
- Adding two negative numbers gives a positive result.

It indicates that the result cannot be represented correctly in signed form.

20. What is the purpose of the OFFSET operator?

A. The OFFSET operator is used to get the memory address of a variable.

Instead of getting the value stored in a variable `OFFSET` gives us its location in memory.

For example:

- If we write `mov eax, OFFSET myVar`,
- EAX will store the address of `myVar`, not its value.

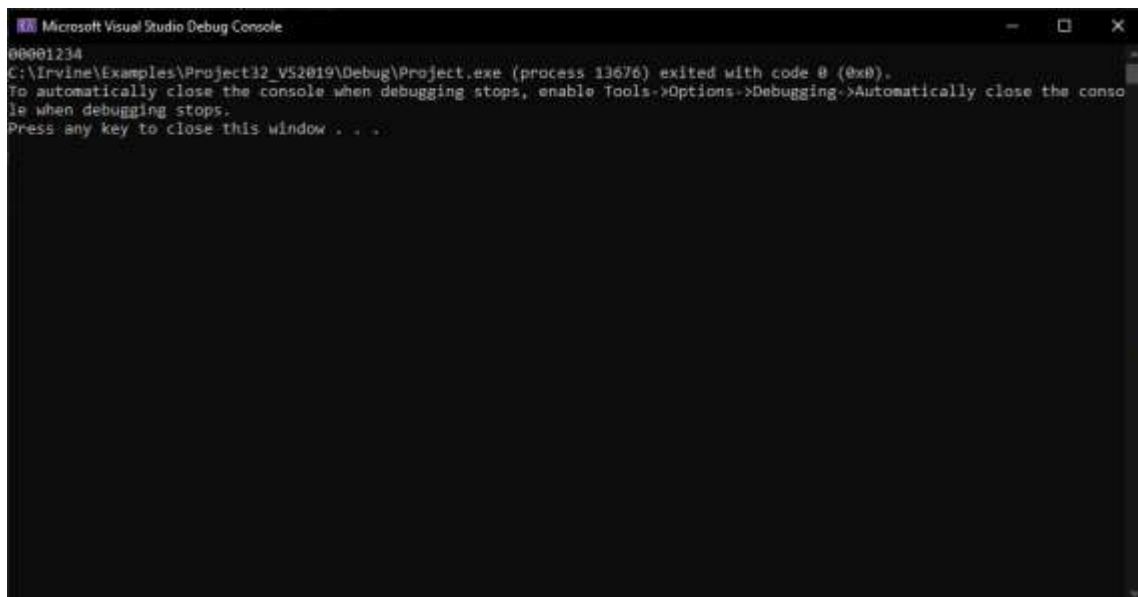
Part B: Short Programming Questions

1.

```
mov ax,1234h mov
bx,ax
```

Find: BX

Answer: BX = 1234h

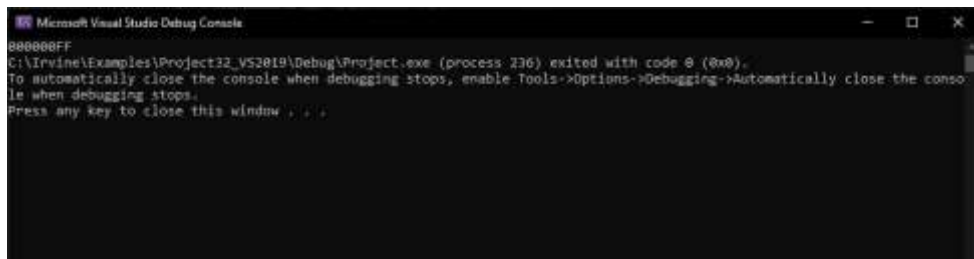


2.

```
mov al,0FFh mov
ah,00h
```

Find: AX

Answer: AX = 00FFh



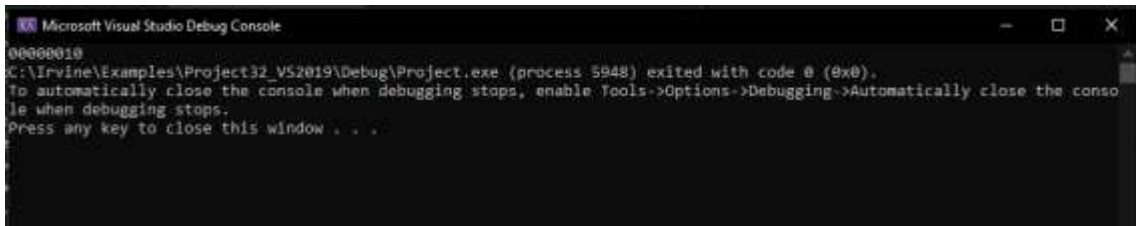
```
Microsoft Visual Studio Debug Console
800000FF
C:\Irvine\Examples\Project32_V52019\Debug\Project.exe (process 236) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .
```

3.

```
.data
val BYTE10h
.code
mov al,val
```

Find: AL

Answer: 10h



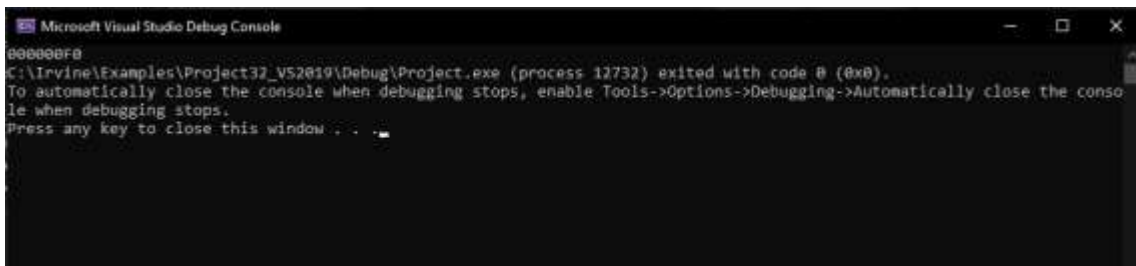
```
Microsoft Visual Studio Debug Console
00000010
C:\Irvine\Examples\Project32_V52019\Debug\Project.exe (process 5948) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .
```

4.

```
mov ax,0 mov
al,0F0h
```

Find: AX

Answer: 00F0h



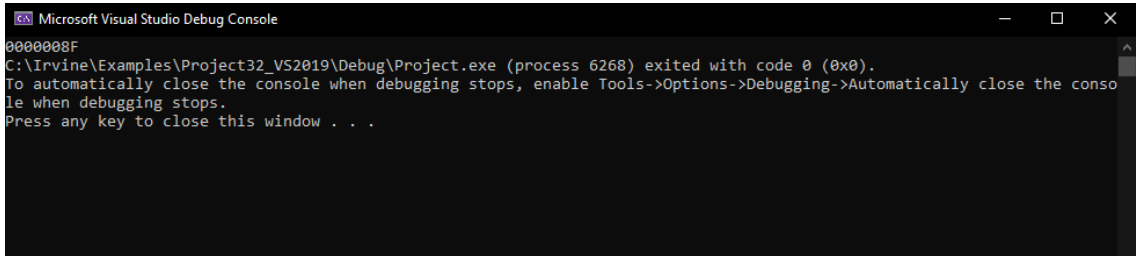
```
Microsoft Visual Studio Debug Console
000000F0
C:\Irvine\Examples\Project32_V52019\Debug\Project.exe (process 12732) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .
```

5.

```
mov bl,10001111b
movzx ax,bl
```

Find: AX

Answer: ____008Fh____



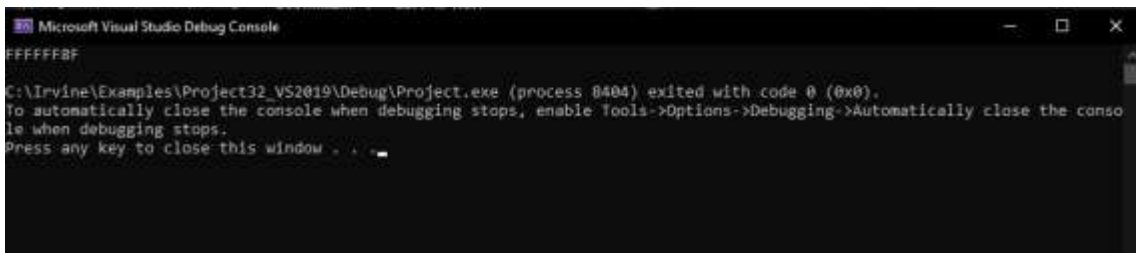
```
Microsoft Visual Studio Debug Console
0000008F
C:\Irvine\Examples\Project32_VS2019\Debug\Project.exe (process 6268) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .
```

6.

```
mov bl,10001111b
movsx ax,bl
```

Find: AX

Answer: ____AX = FF8Fh____



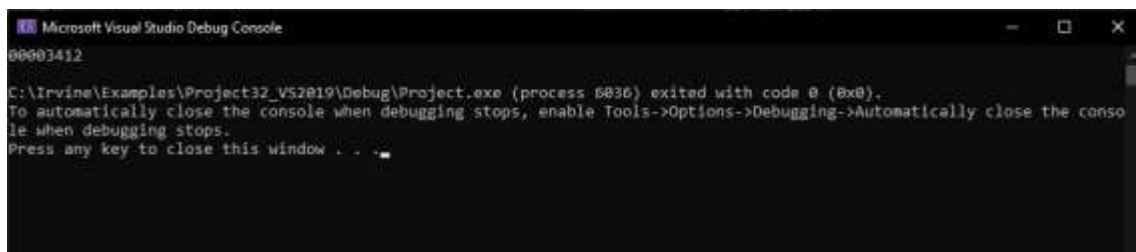
```
Microsoft Visual Studio Debug Console
FFFFFFFF
C:\Irvine\Examples\Project32_VS2019\Debug\Project.exe (process 8404) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .
```

7.

```
mov ax,1234h
xchg al,ah
```

Find: AX

Answer: ____AX = 3412h____



```
Microsoft Visual Studio Debug Console
00003412
C:\Irvine\Examples\Project32_VS2019\Debug\Project.exe (process 8036) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .
```


8.

.data

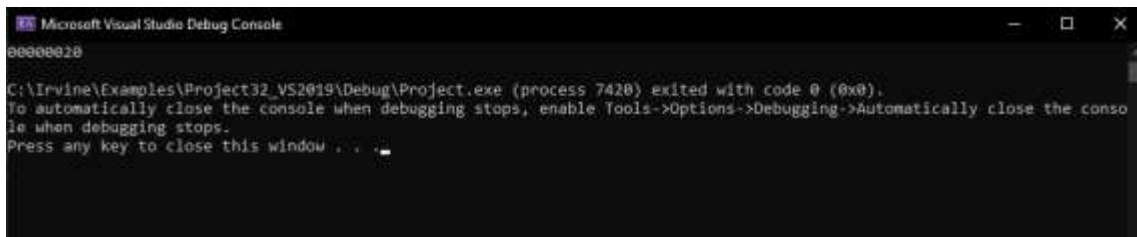
arrayB BYTE 10h,20h,30h

.code mov

al,arrayB+1

Find: AL

Answer: _____ AL = 20h _____



```
Microsoft Visual Studio Debug Console
00000020
C:\Irvine\Examples\Project32_V52019\Debug\Project.exe (process 7420) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .
```

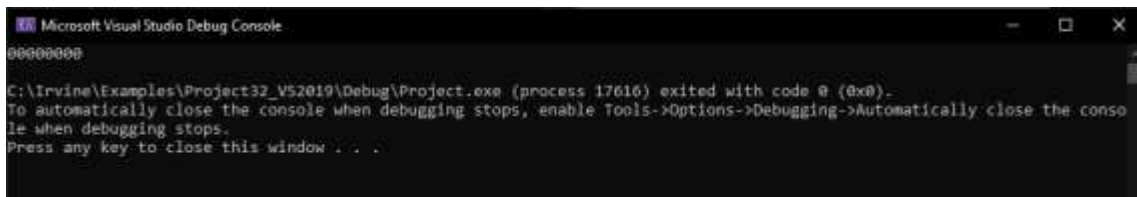
9.

mov al,0FFh add

al,1

Find: AL, CF, ZF

Answer: _____ AL = 00h , CF = 1 , ZF = 1 _____



```
Microsoft Visual Studio Debug Console
00000000
C:\Irvine\Examples\Project32_V52019\Debug\Project.exe (process 17616) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .
```

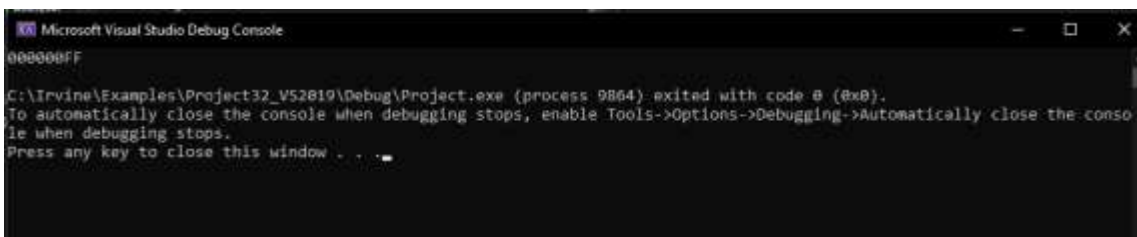
10.

mov al,0

sub al,1

Find: AL, CF, SF

Answer: _____ AL = FFh , CF = 1 , SF = 1 _____



```
Microsoft Visual Studio Debug Console
000000FF
C:\Irvine\Examples\Project32_V52019\Debug\Project.exe (process 9864) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .
```

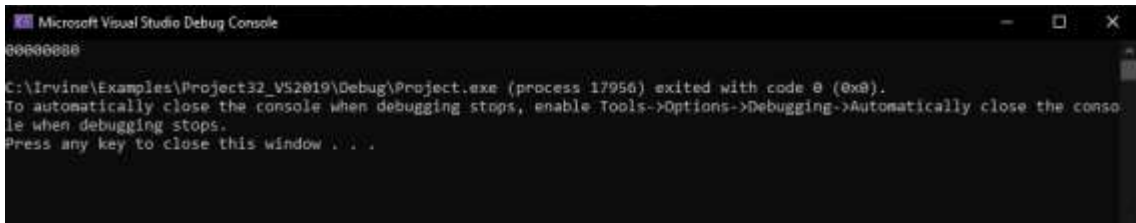
11.

mov al,7Fh

add al,1

Find: AL, OF

Answer: _____AL = 80h , OF = 1_____



```
Microsoft Visual Studio Debug Console
00000080
C:\Irvine\Examples\Project32_V52019\Debug\Project.exe (process 17956) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .
```

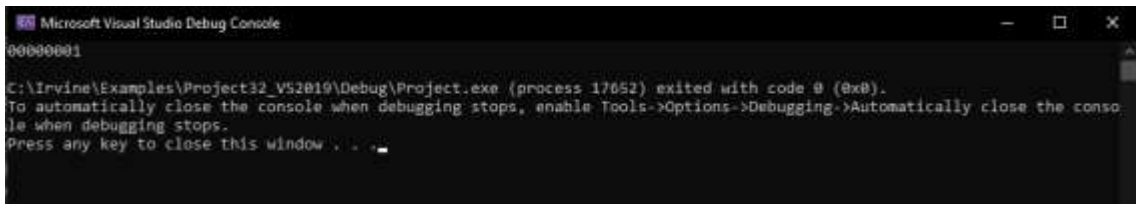
12.

mov al,-1

neg al

Find: AL

Answer: _____AL = 01h_____



```
Microsoft Visual Studio Debug Console
00000001
C:\Irvine\Examples\Project32_V52019\Debug\Project.exe (process 17652) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .
```

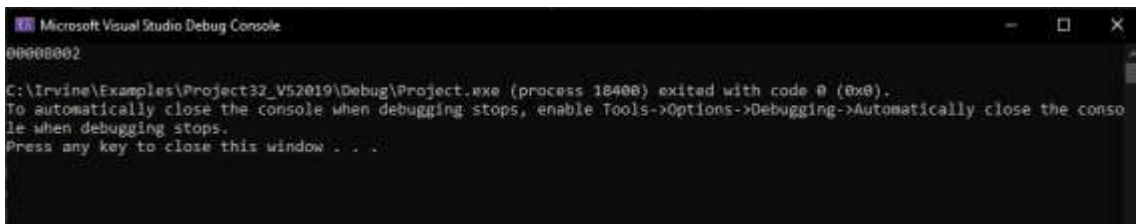
13.

mov ax,8000h

add ax,2

Find: AX, OF

Answer: _____AX = 8002h , OF = 0_____



```
Microsoft Visual Studio Debug Console
00000002
C:\Irvine\Examples\Project32_V52019\Debug\Project.exe (process 18400) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .
```

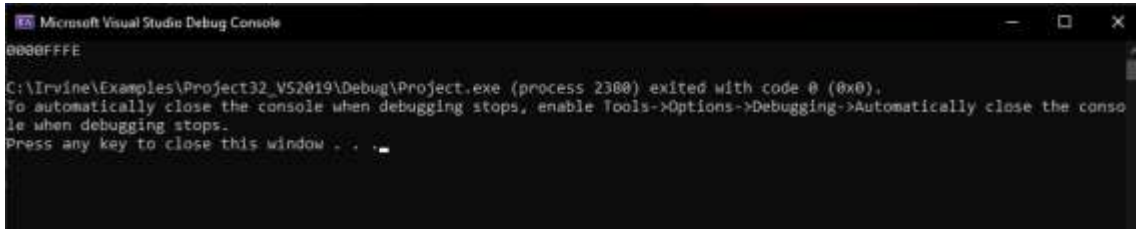
14.

mov ax,0 sub

ax,2

Find: AX, CF

Answer: _____ AX = FFFEh , CF = 1 _____



```
Microsoft Visual Studio Debug Console
0000FFFF
C:\Irvine\Examples\Project32_VS2019\Debug\Project.exe (process 2380) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .
```

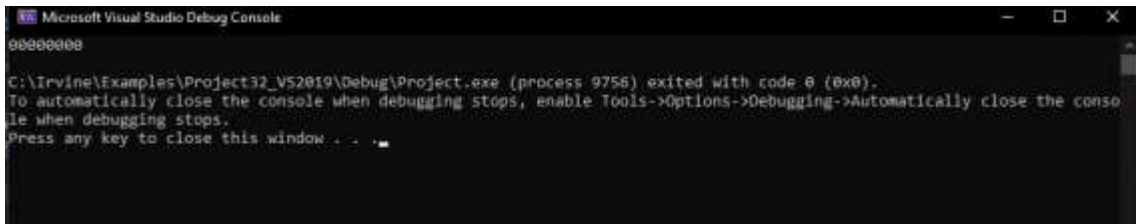
15.

mov al,5 sub

al,5

Find: AL, ZF

Answer: _____ AL = 00h , ZF = 1 _____



```
Microsoft Visual Studio Debug Console
00000000
C:\Irvine\Examples\Project32_VS2019\Debug\Project.exe (process 9756) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .
```

16.

.data

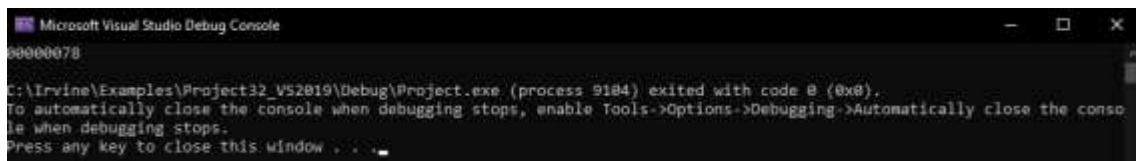
myDouble DWORD 12345678h

.code

mov al,BYTE PTR myDouble

Find: AL

Answer: _____ AL = 78h _____



```
Microsoft Visual Studio Debug Console
00000078
C:\Irvine\Examples\Project32_VS2019\Debug\Project.exe (process 9184) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .
```

17.

mov ax,WORD PTR myDouble

Find: AX

Answer: _____ AX = 5678h _____

18.

```
.data varB BYTE  
12h,34h,56h,78h  
.code mov ax,WORD PTR  
[varB+2]  
Find: AX
```

Answer: _____ AX = 7856h _____

19.

```
.data  
arrayW WORD 1000h,2000h,3000h  
.code mov  
ax,[arrayW+4]
```

Find: AX

Answer: _____ AX = 3000h _____

Part C: Programming Problems

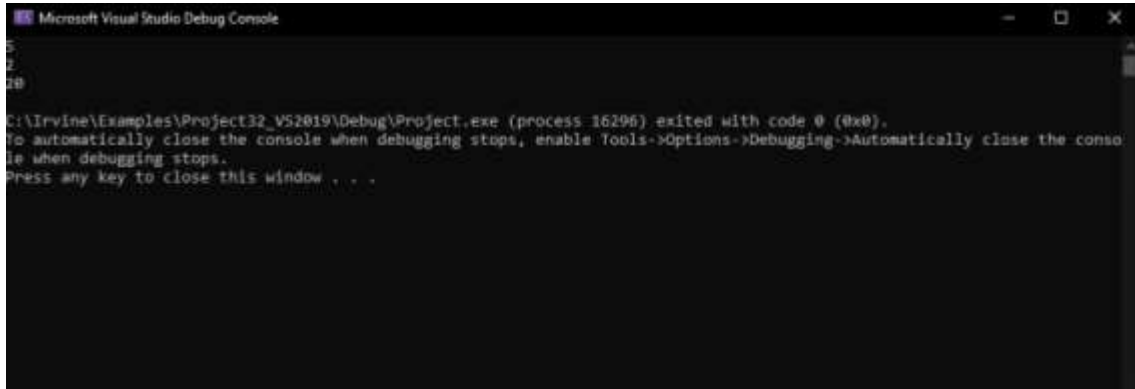
Problem 1: Data Definition Practice

Write a MASM program that declares:
- A BYTE variable initialized to 100

- A WORD variable initialized to 5000h
- A DWORD array of 5 elements (uninitialized)
- A string: "Assembly Language",0

Calculate and display:

- LENGTHOF array
- TYPE of WORD variable
- SIZEOF array

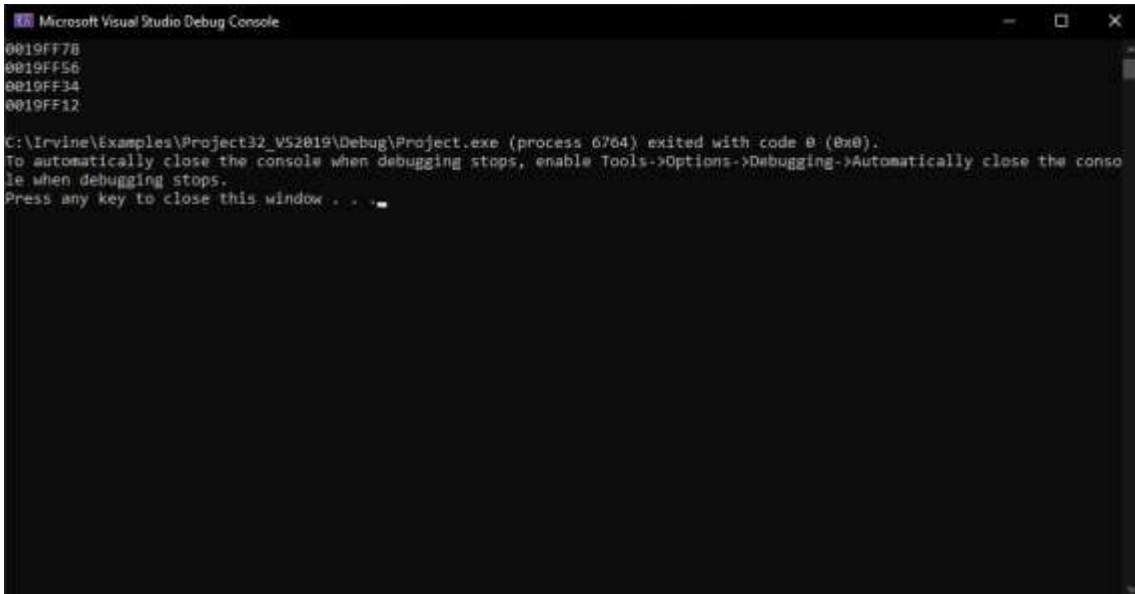


```
Microsoft Visual Studio Debug Console
5
2
20
C:\Irvine\Examples\Project32_VS2019\Debug\Project.exe (process 16296) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .
```

Problem 2: Little Endian Verification

Declare: myVal DWORD 12345678h

Write a program that uses BYTE PTR to display each byte separately. Explain the output based on Little Endian order.



```
Microsoft Visual Studio Debug Console
0019FF78
0019FF56
0019FF34
0019FF12
C:\Irvine\Examples\Project32_VS2019\Debug\Project.exe (process 6764) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .
```

Problem 3: Rearranging

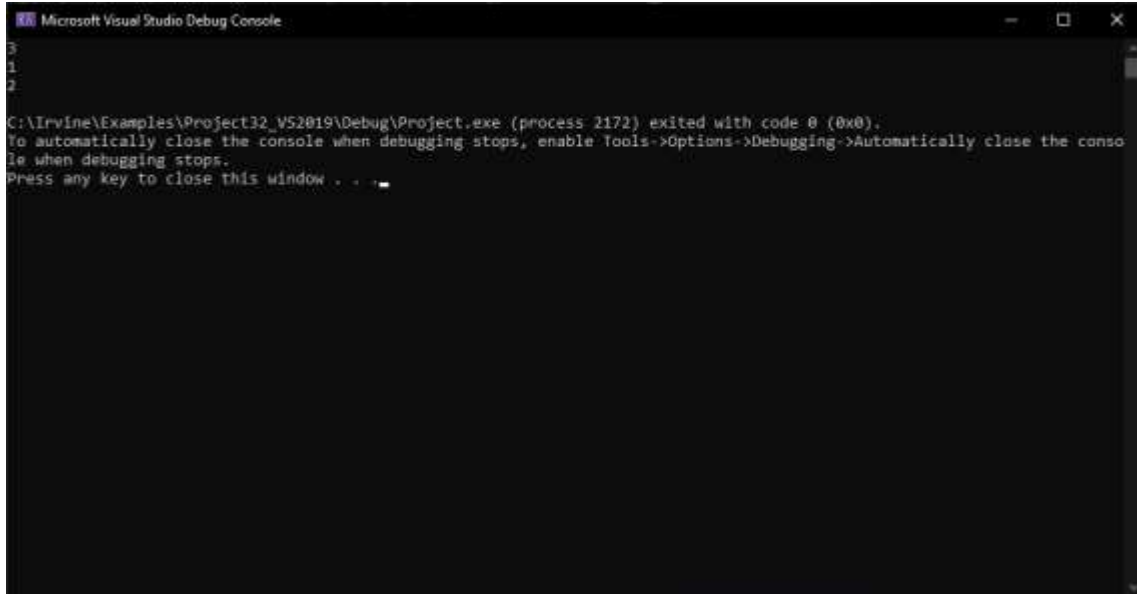
Given: arrayD DWORD 1,2,3

Rearrange values into: 3,1,2 Use
MOV and XCHG.

Do not create another array.

Use direct offset addressing.

Update the code for byte type array.



```
Microsoft Visual Studio Debug Console
3
1
2
C:\Irvine\Examples\Project32_V52019\Debug\Project.exe (process 2172) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .
```